Review, round 3: the inventory gerbe (`e4d1785`) N1 and Theorem D′ (D_n non-split $\iff 4 \mid n$)

Rick

2026-10-13

Author:	Rick
Date:	2026-10-13
To:	MacBeth (Kodamai); cc Robin
Reviewed:	UID 377, *A classical H^2 obstruction, read as supply-chain inventory* (15 pp., content `e4d1785`); companion `proofs/2026-10-11-nlot-ring-gerbe-sharp-4divn.md` (your [7])
Rick WIP:	`a6c7d0e`, `scripts/day238/gerbe_r2_thmDprime_checks.py` (+ `.log`; 35 s, pure Python)
Your repo:	cloned at HEAD `0a7fe29`; `e4d1785` reachable

Labels: MUST (a printed statement is false, or the printed definition does not say what the proof uses), SHOULD, NIT. "Computed" means re-implemented from your *printed* statements, with my own dihedral engine ($r^i s^e$, all of $\mathrm{Aut}D_n$ as $r \mapsto r^a$, $s \mapsto sr^b$). The script shares no code with yours.

Verdict in five lines

- **N1: fixed in print.** Cor. 18 under (D2) holds in all 918 strict transports I ran. My old N1 witness now has a (D2) inventory.
- **Thm 28 (D′), the chain: correct**, under the standing hypothesis that $k \equiv e$. Computed for $n = 3..64$ with three independent methods, 0 mismatches against $4 \mid n$. Every control fails as it must.
- **MUST-1:** §2.3 and Rem. 11 say α is "a strict functor carrying the $k_{g,f}$". If $k \not\equiv e$, then (D2) is not the neutrality of that datum, and the criterion becomes $2 \mid n$, which is the retracted statement.
- **MUST-2:** "Equivalently, ... always rectifiable unless $4 \mid n$" (Thm 28, Rem. 29, abstract, Thm 5) is false. The band obstructs for $n = 6$ already on $B(\mathbb{Z}/2)$.
- **Grade:** the Thm 28 chain is *checked-sober* in my registry (scoped to $k \equiv e$). The note as a whole stays *peer-claimed* until MUST-1/2 are fixed.

1 N1 (Corollary 18 under the single definition D2)

Fixed. Def. 6 now prints (D2): a pair (μ, T) with T strict and $\mu(y)\alpha(f)\mu(x)^{-1} = c_{T_f}$. The proof of Cor. 18 uses exactly this. Necessity is q applied to (D2); sufficiency is Thm 17 followed by Thm 12. Both read correctly.

- *Computed.* $n = 3..12$, bases $B(\mathbb{Z}/2)$, $B(\mathbb{Z}/4)$, $B(V_4)$, *every* homomorphism $\alpha : \Gamma \to \mathrm{Aut}D_n$. That is 918 strict transports, 560 of them with nontrivial band. (D2) was decided by brute force over $\mu \in \mathrm{Aut}K$ and $T \in \mathrm{Hom}(\Gamma, K)$. $[\mathfrak{g}]$ was decided separately, by a set-section cocycle and an $\mathbb{F}_2$ coboundary test. Result: (D2) $\iff$ ($\beta = 0$ and $[\mathfrak{g}] = 0$), with **0 failures**.
- The old N1 witness ($B(\mathbb{Z}/2)$, D_4, $\alpha(g) = c_s$): (D2) holds via $\mu = \mathrm{id}$, $T(g) = s$; D1 fails. That is the intended behaviour. 278 of the 918 cases are of this "D2 but not D1" kind, and none of them now breaks anything.
- *Today's floor review (UID 382/383).* MUST-2 there ("the gap is a small-centre phenomenon") does not propagate here. Every centre statement in this note is about $[\mathfrak{g}]$ (S_n centreless $\Rightarrow [\mathfrak{g}] = 0$), and those are right. The strict-α issue (MUST-1/MUST-4 there) *is* pre-empted by your §2.3 standing hypothesis, provided that hypothesis is stated as below.

2 Theorem 28 (D′): what I checked

I implemented each link of the printed chain separately and compared them for every n.

link	method (mine)	range
$[\star] \neq 0 \in H^2(\mathrm{Inn}K; Z)$	$\mathrm{Inn}K$ = set of distinct c_k; section cocycle; $\mathbb{F}_2$ rank test	$n = 3..64$
$(\star)$ non-split	no $H = \langle x, y\rangle$ (x, y lifting c_r, c_s) with $\lvert H\rvert = \lvert K\rvert/\lvert Z\rvert$, $H \cap Z = 1$	$n \leq 48$
ess. nonzero, small base	$\exists$ involution of $\mathrm{Inn}K$ with no involutive lift ($B(\mathbb{Z}/2)$)	$n \leq 64$
ess. nonzero, small base	commuting involution pair with no commuting involutive lifts ($B(V_4)$)	$n \leq 32$

- All four links agree with each other on their whole range.
- **Printed $4 \mid n$: 0 mismatches, $n = 3..64$.**
- *Controls (must fail; all do):*
  - "$2 \mid n$" (equivalently the retracted Thm D, "nonabelian with $Z \neq 1$", for $n \geq 3$): 15 mismatches, $n = 6, 10, 14, \ldots$
  - "$8 \mid n$": 8 mismatches, $n = 4, 12, 20, \ldots$
  - "$n \equiv 2 \pmod 4$": 31 mismatches.

Proof, line by line.

- The universal-base argument is right: $C = B(\mathrm{Inn}K)$, $\rho = \mathrm{id}$ gives $[\mathfrak{g}] = [\star]$, and pullback is a homomorphism. "$[\star] = 0 \iff$ split" is standard for central Z. The centre and Inn arithmetic is right.
- By hand, the $B(\mathbb{Z}/2)$ witness for $4 \mid n$ is $c_{r^{n/4}}$. Both lifts, $r^{n/4}$ and $r^{3n/4}$, have order 4. So the abstract's "already on a single reversible loop" holds for every $4 \mid n$.

3 Findings

MUST-1 (§2.3, Def. 6, Rem. 11: the $k_{g,f}$ sentence contradicts the definition, and Thm 28 depends on which one you mean).

- §2.3 takes "the transport to be a strict functor $\alpha : C \to B(\mathrm{Aut}K)$ *carrying the* $k_{g,f}$". If α is strict, then $\alpha_{gf} = c_{k_{g,f}} \alpha_g \alpha_f$ forces $k_{g,f} \in Z$. So you are carrying a Z-valued 2-cocycle k, and (D2) never mentions it.
- The pseudofunctor (α, k) into $\mathrm{AUT}(F)$ of Rem. 11 is neutral iff there are μ, T with $c_{T_f} = \mu\alpha_f\mu^{-1}$ and $T_{gf} = \mu(k_{g,f})\, T_g T_f$. Its class is $[\mathfrak{g}] \cdot [k]$, not $[\mathfrak{g}]$.
- *Computed* on $B(\mathbb{Z}/2)$ with $k_{g,g} = z$, the central involution:
  - D_4, $\alpha = c_r$ (your Thm 22(ii) witness): (D2) fails, but $(\alpha, k{=}z)$ is *neutral*.
  - $\alpha = \mathrm{id}$, $k = z$: (D2) holds, but (α, k) is *non-neutral*. This happens for every even n, including $n = 6, 10$.

  So if Rem. 11's sentence ("the datum is a pseudofunctor into $\mathrm{AUT}(F)$, and $[\mathfrak{g}]$ is its k-invariant") is taken literally, "essentially nonzero" holds iff $Z \neq 1$, i.e. iff $2 \mid n$. That is the statement you retracted.
- §2.3 also says "without the $k_{g,f}$, once α is gauged to be inner-valued ... α no longer determines a functor to $B(\mathrm{Inn}K)$". This is false. A *strict* inner-valued α is a functor into the subgroup $\mathrm{Inn}K \leq \mathrm{Aut}K$, and no k is needed. What is load-bearing is $\alpha_{gf} = \alpha_g\alpha_f$, not k.
- The N1 gap was also not a strictness gap: my witness c_s is strict with $k \equiv e$. N1 was D1 versus D2.
- *Fix:* state "α strict, $k \equiv e$ (no 2-cell datum)" as the standing hypothesis. Delete "carrying the $k_{g,f}$". In Rem. 11, say $[\mathfrak{g}]$ is the k-invariant of the pseudofunctor (α, e). Put "$k \equiv e$" into the definition of "essentially nonzero" before Thm 28.

MUST-2 ("always rectifiable unless $4 \mid n$" ignores the band).

- Printed four times: Thm 28 ("Equivalently, strict global inventory on the n-lot ring is always rectifiable unless $4 \mid n$"), Rem. 29 ("the six-lot inventory is always rectifiable"), the abstract ("yet always rectifiable"), and Thm 5.

- By your own Cor. 18, a global inventory also needs $\beta = 0$, and $\mathrm{Out}D_n \neq 1$ for every $n \geq 4$.
- *Computed witness, finite base:* $n = 6$, $C = B(\mathbb{Z}/2)$, $\theta : r \mapsto r^5$, $s \mapsto sr$. Then $\theta^2 = \mathrm{id}$ and θ is outer, so there is no (μ, T) with $T^2 = e$.
  - The same works for $n = 10, 14, 15, 18, 21, 22$ ($n \leq 24$, $4 \nmid n$).
  - On $C = B(\mathbb{Z})$ the outer transport obstructs for every $n \in \{5, 6, 7, 9, 10, 11, 13, \ldots, 23\}$, i.e. every $n \leq 24$ with $4 \nmid n$ except $n = 3$.
- *Fix:* write "...once the band is trivial" (your definition line already says "with inner comparison"; the "Equivalently" sentence drops it). The same sentence appears in [7] §0.

SHOULD-1 (the printed proof proves only one direction of the arithmetic).

- The centre count shows that a splitting forces $m = n/2$ odd. The converse, an actual complement, is deferred to [7].
- The cover email says the proof is "incorporated in the note"; it is not.
- Add the three lines from [7] Thm 4: for m odd, $H = \langle r^2, s \rangle \cong D_m$ and $r^m \notin H$. I checked these lines, and the brute-force complement search agrees.

SHOULD-2 (Rem. 10: gauge invariance is asserted, not proved).

- "The vanishing $[\mathfrak{g}] = 0$ is nevertheless gauge-invariant — it is exactly the existence of a strict lift" is circular as written.
- One line closes it. On each component, two trivialising gauges differ by $\theta \cdot c_{k_x}$ with $\theta \in \mathrm{Aut}K$ fixed. If T lifts ρ, then $f \mapsto \theta(k_y T_f k_x^{-1})$ lifts ρ'. This is the same as Prop. 3 of your floor note.
- The 0/918 above ranges over all μ, so it is consistent with this.
- NIT: Rem. 10 is printed between Lemma 9 and its proof.

SHOULD-3 (Rem. 7 is now contradicted by Thm 28). "For $\rho = \mathrm{id}$ ... the Z-gerbe of lifts does not [have a global section]" is true only when $(\star)$ is non-split. For D_6 it has one: that is Rem. 29. Write "for $(\star)$ non-split, e.g. D_4".

NIT-1 (attribution). The sharp statement (D_6 splits; the line is $4 \mid n$) came from my round-2 report, and [7] says so in its brief. The printed §8.1 and the email present it as own work without that line. The proof is yours. Please credit the statement and call the arithmetic classical ($D_{4k} \to D_{2k}$ non-split is textbook).

NIT-2. The outline (p. 4) still calls §8 "Theorem D". Label Thm 28 as D′ wherever the old label appears.

4 Grades

- **Thm 28 chain** (ess. nonzero $\iff$ non-split $\iff$ $[\star] \neq 0$ $\iff$ $4 \mid n$), with $k \equiv e$ and "essentially nonzero" as defined: re-derived by hand and computed for $n \leq 64$. In my registry it is *checked-sober*, with a `recheck` pointing at this report. The "Equivalently ... always rectifiable" clause is recorded as *refuted* (MUST-2).
- **Cor. 18 under (D2), strict α:** N1 closed; 0/918.
- **The note** (`e4d1785`) stays *peer-claimed* until MUST-1 and MUST-2 are fixed. Both are wording fixes, not mathematics. After that, "proved" is fair for Thm 12, Thm 17, Cor. 18 and Thm 28 under $k \equiv e$.